

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Networks

COURSE CODE: CSA0706

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I __N.Umasri_(192424240)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

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Short Answer Test

1. A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Answer:

Given:

- File size = **150 MB** = $150 \times 10^6 = 150,000,000$ bytes
- Segment size = **1500 bytes**
- Frame overhead = **40 bytes**
- Bandwidth = **50 Mbps**

Calculations:

1. **Number of segments**
= $150,000,000 \div 1500$
= **100,000 segments**
2. **Number of frames**
= **100,000 frames** (1 segment = 1 frame)
3. **Total overhead**
= $100,000 \times 40$
= **4,000,000 bytes (4 MB)**

4. **Total data transmitted**

$$= 150,000,000 + 4,000,000$$

$$= \mathbf{154,000,000 \text{ bytes}}$$

5. **Transmission time**

$$= (154,000,000 \times 8) \div 50,000,000$$

$$= \mathbf{24.64 \text{ seconds}}$$

Explanation:

- **Transport Layer:** Divides the file into segments, ensures reliable delivery using acknowledgements, sequencing, and retransmission.
- **Data Link Layer:** Converts segments into frames, performs error detection using CRC, and ensures reliable communication between directly connected devices.

2. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

Given:

- Window size = **6 frames**
- Total frames = **48**

Calculations:

1. **Transmission windows**

$$= 48 \div 6$$

$$= \mathbf{8 \text{ windows}}$$

2. **Retransmissions**

$$= 1 \text{ frame lost per window}$$

$$= \mathbf{8 \text{ retransmissions}}$$

Explanation:

Flow control prevents the sender from sending data faster than the receiver can process it, reducing congestion and improving communication efficiency.

3. A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Answer:

Given:

- Packet size = **12,000 bytes**
- MTU = **1500 bytes**
- IP header = **20 bytes**

Calculations:

Maximum data per fragment

$$= 1500 - 20$$

$$= \mathbf{1480 \text{ bytes}}$$

Number of fragments
= Ceiling($12,000 \div 1480$)
= **9 fragments**

Header overhead
= 9×20
= **180 bytes**

Explanation:

The Network Layer fragments packets when required and uses fragment offsets, identification numbers, and flags so that the destination can correctly reassemble the original packet.

4. A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Answer:

Calculations:

- Number of windows = $48 \div 6 = 8$
- Retransmissions = **8**

Explanation:

Flow control ensures efficient communication by preventing receiver overload and reducing unnecessary retransmissions.

5. A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Answer:

Given:

- File size = **600 MB**
- Checkpoint every **60 MB**
- Failure after **390 MB**

Calculations:

Checkpoints created

$$= 390 \div 60$$

$$= \mathbf{6 \text{ checkpoints}}$$

Last checkpoint

$$= \mathbf{360 \text{ MB}}$$

Data to retransmit

$$= 390 - 360$$

$$= \mathbf{30 \text{ MB}}$$

Explanation:

Synchronization allows communication to resume from the last checkpoint instead of restarting from the beginning, saving time and bandwidth.

6. Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Answer:

Given:

- Original file = **900 MB**
- Compression = **40%**
- Encryption overhead = **5%**

Calculations:

Compressed size

$$= 900 \times 60\%$$

$$= \mathbf{540 \text{ MB}}$$

Encrypted size

$$= 540 \times 1.05$$

$$= \mathbf{567 \text{ MB}}$$

Reduction

$$= ((900 - 567) \div 900) \times 100$$

$$= \mathbf{37\% \text{ reduction}}$$

Explanation:

The Presentation Layer compresses data to reduce transmission time and encrypts data to provide confidentiality and security.

7. An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Answer:

Given:

- Data = **3 GB = 3000 MB**
- Request size = **3 MB**
- Throughput = **120 Mbps**

Calculations:

Requests

$$= 3000 \div 3$$

$$= \mathbf{1000 \text{ requests}}$$

Transmission time

$$3 \text{ GB} = 24,000 \text{ Mb}$$

Time

$$= 24,000 \div 120$$

$$= \mathbf{200 \text{ seconds}}$$

Explanation:

The Application Layer provides services such as file transfer, user authentication, error reporting, and communication between applications.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Answer:

Calculations:

$$\begin{aligned} \text{Processing delay} \\ &= 4 + 3 + 5 + 2 + 1 \\ &= \mathbf{15 \text{ ms}} \end{aligned}$$

$$\begin{aligned} \text{Total delay} \\ &= 15 + 18 + 12 \\ &= \mathbf{45 \text{ ms}} \end{aligned}$$

Explanation:

- Application Layer provides user services.
- Transport Layer ensures reliable delivery.
- Network Layer performs routing.
- Data Link Layer handles framing and error detection.
- Physical Layer transmits bits over the medium.

9. A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Answer:

Given:

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Overhead = **80 bytes**

Useful data per frame

$$= 1600 - 80$$

$$= \mathbf{1520 \text{ bytes}}$$

Calculations:

Frames required

$$= \text{Ceiling}(80,000,000 \div 1520)$$

$$= \mathbf{52,632 \text{ frames}}$$

Total overhead

$$= 52,632 \times 80$$

$$= \mathbf{4,210,560 \text{ bytes}}$$

Efficiency

$$= (1520 \div 1600) \times 100$$

$$= \mathbf{95\%}$$

Explanation:

Protocol overhead ensures reliable communication, addressing, and error detection, but excessive overhead reduces available bandwidth and overall efficiency.

10. A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Answer:

Given:

- Original file = **240 MB**
- Compression = **30%**
- Segment size = **1400 bytes**
- Frame overhead = **60 bytes**

Calculations:

Compressed size

$$= 240 \times 70\%$$

$$= \mathbf{168 \text{ MB}}$$

$$= \mathbf{168,000,000 \text{ bytes}}$$

Segments

$$= 168,000,000 \div 1400$$

$$= \mathbf{120,000 \text{ segments}}$$

Protocol overhead

$$= 120,000 \times 60$$

$$= \mathbf{7,200,000 \text{ bytes}}$$

Final data transmitted
= 168,000,000 + 7,200,000
= **175,200,000 bytes (175.2 MB)**

Explanation:

- **Presentation Layer:** Compresses and encrypts data.
- **Transport Layer:** Segments data and ensures reliable delivery.
- **Network Layer:** Routes packets.
- **Data Link Layer:** Adds frames and performs error detection.
- **Physical Layer:** Transmits the data through the communication medium.